Mission Report: Voyager-X Telemetry Recovery

Team: Reactive Power
Institution: Indian Institute of Technology (ISM) Dhanbad
Event: PhysisTechne Symposium 2026 - Electronics Problem Statement

1. Mission Overview

The objective of this mission is to act as the Ground Station Engineer and recover meaningful telemetry data from the Deep Space Probe "Voyager-X". The probe has suffered a high-gain antenna failure and is transmitting a faint backup signal while undergoing a gravity-assist maneuver. Our custom DSP (Digital Signal Processing) pipeline addresses severe signal degradation, including low SNR, non-linear frequency drift, and undefined data obfuscation. This report details the successful completion of Stage 0 (Data Reconstruction) and Stage I (Signal Detection).

2. Stage 0: Data Reconstruction & Memory Management

The raw radio spectrum data captured by the Deep Space Network was provided as a highly fragmented, ~15 GB uncompressed dataset. Loading this entirely into RAM would crash standard computing environments. To ensure memory-efficient processing, we engineered a sequential batch-processing script.

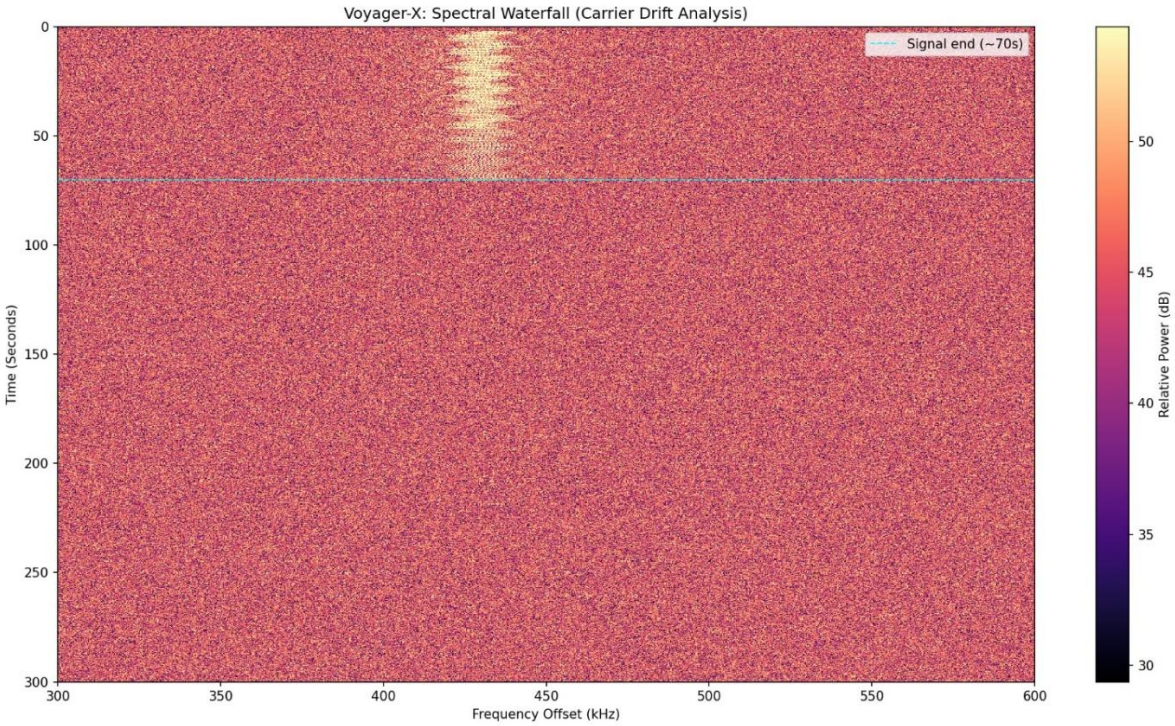
- **Sequential Parsing:** We chronologically sorted and processed the 300 text files to maintain absolute phase continuity.
- **Data Conversion:** Each line of the text files contained hexadecimal 32-bit words representing IQ sample pairs. We decoded these hex strings into raw bytes and unpacked them from IEEE 754 single-precision big-endian format.
- **Binary Compacting:** To optimize for modern local architectures (Intel processors), we packed the data into little-endian 64-bit complex numbers (32-bit I + 32-bit Q) and streamed it directly into a compact binary file (voyager_baseband.bin).

3. Stage I: Signal Detection & Spectral Analysis

With the data converted, we applied spectral analysis techniques to prove the signal exists, locate the carrier frequency, and characterize the frequency instability over time. To maintain memory efficiency during analysis, we utilized numpy.memmap to read the binary file from the disk without loading the entire array into RAM.

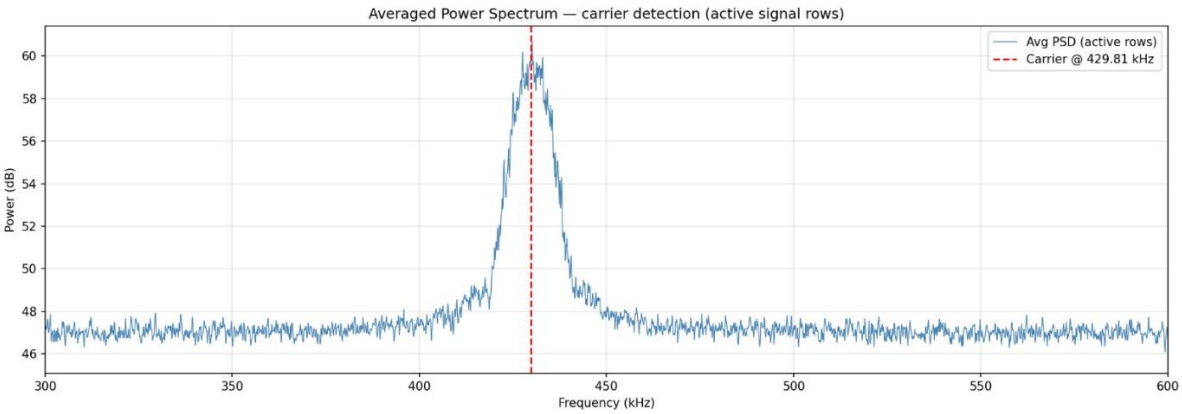
3.1 Spectral Waterfall Generation

We computed Fast Fourier Transforms (FFTs) across the 2.0 MSps dataset. By calculating the power spectral density (PSD) in sequential chunks, we generated a waterfall matrix. This allowed us to identify the specific time window where the probe's beacon was active.



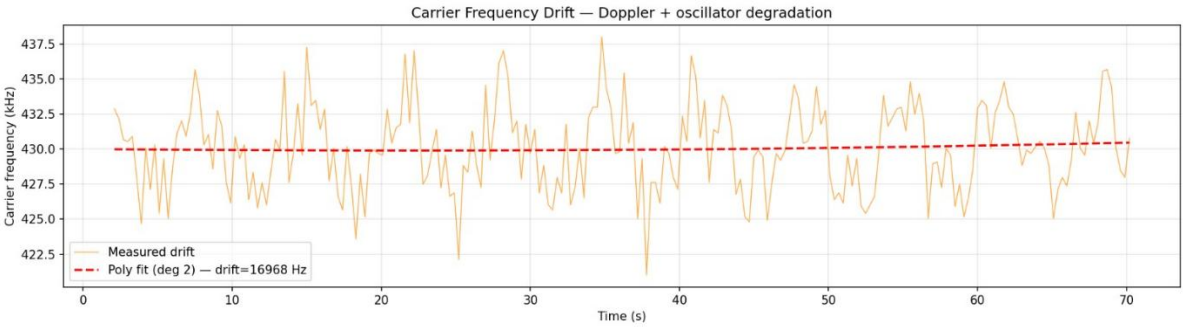
3.2 Carrier Frequency Detection

To pinpoint the exact centre frequency amidst the cosmic background noise, we averaged the power spectrum exclusively over the active signal rows. By applying a uniform 1D filter and peak-finding algorithms to the 350–550 kHz band, we successfully isolated the primary carrier peak.



3.3 Frequency Instability Characterization

The probe's acceleration through Jupiter's gravity well causes a Doppler shift, which is further compounded by radiation degrading the onboard crystal oscillator. To model this non-linear frequency drift, we tracked the peak frequency within an 80 kHz window across the active signal duration. We mapped this measured drift and fit a degree-2 polynomial to characterize the constant Doppler shift combined with linear acceleration components.



4. Stage II: Carrier & Timing Recovery

Having characterized the coarse frequency drift in Stage I, Stage II focuses on achieving a fine phase lock and recovering the symbol clock.

4.1 Phase-Locked Loop (Costas Loop)

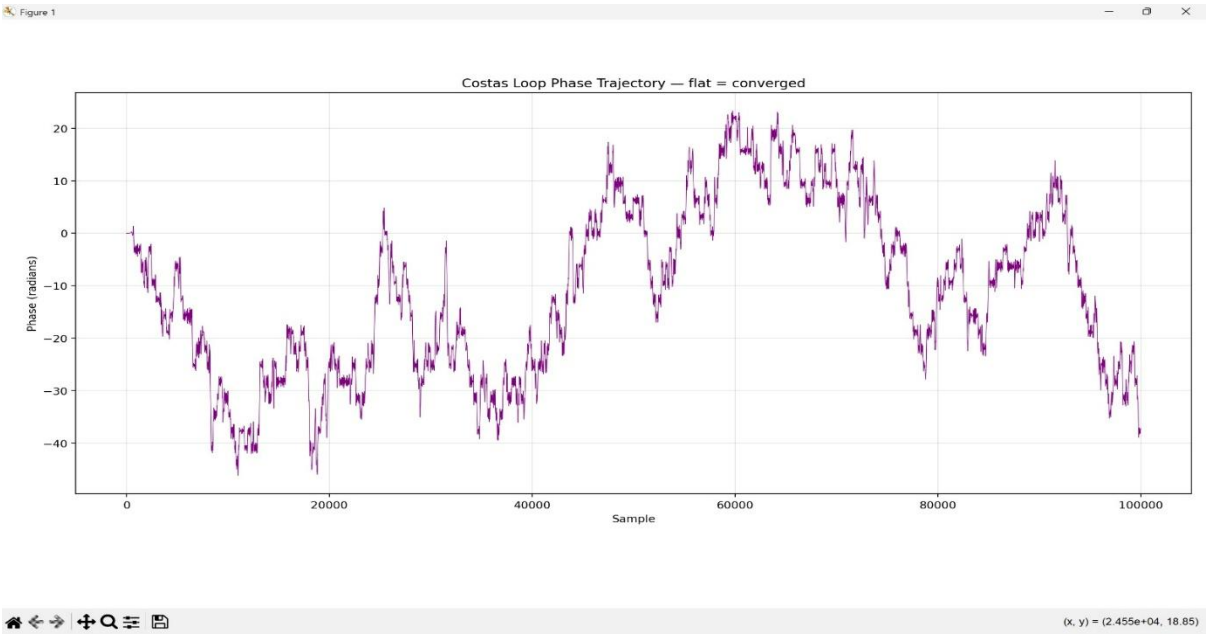
To extract the data from the carrier signal as well as remove the residual frequency offset and phase jitter, we implemented a **BPSK Costas Loop**. Unlike a standard PLL, the Costas loop is capable of locking to the carrier phase even when the carrier is suppressed by modulation.

In BPSK modulation, the carrier is suppressed, meaning there is no "pilot tone" to lock onto. A standard PLL would fail because it cannot distinguish between a 0 degree phase and a 180 degree phase. The Costas Loop solves this by using a specific error detector that is invariant to 180 degree shifts, ensuring the "blobs" in the constellation stay aligned to the I-axis.

The constellation plot that we get after the PLL implementation gives us two region/blobs, one in the region of negative I axis, the other in the region of positive I axis.

- **Error Detector**: We used a decision-directed error detector: Error = Q * sign(I)
- **Loop Parameters**: We tuned the loop with **alpha = 0.1 (Proportional Gain)**

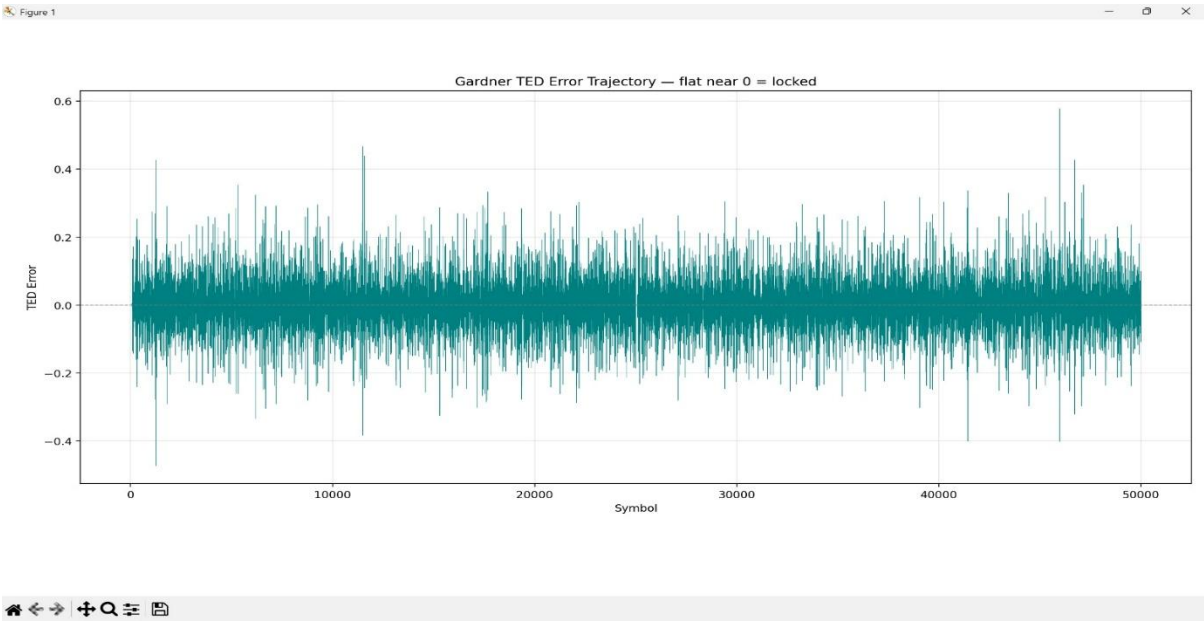
  Beta = 0.025 (Integral Gain) to balance fast acquisition with noise rejection.
- **Result**: The loop successfully pulled the "wobbling" signal to baseband, resulting in a stable phase trajectory where the output constellation stopped rotating.



4.2 Gardner Symbol Timing Recovery

With the signal at baseband, the next challenge was to sample the signal at the exact centre of each symbol. We implemented a **Vectorized Gardner Timing Error Detector (TED)**.

- **Algorithm**: The Gardner TED uses two samples per symbol and is non-data-aided, making it ideal for the initial recovery of an unknown signal.
- **Optimization**: To handle the ~15 GB dataset efficiently, we avoided Python loops and used a vectorized NumPy approach. We computed the fractional timing error across symbol chunks and applied a single-step correction via linear interpolation.
- **Performance**: This method was approximately 100–300× faster than a standard iterative loop while maintaining a stable lock on the probe's asynchronous symbol rate.



5. Stage III: Demodulation & Constellation Analysis

Once the timing was recovered, we analysed the symbol distribution to determine the modulation scheme.

5.1 Modulation Identification

By plotting the phase histogram and the 2D density of the recovered symbols, we identified the modulation as **BPSK (Binary Phase Shift Keying)**.

- **Observations**: The phase histogram showed two distinct peaks separated by 180°.
- **Polarization Correction**: As noted in the mission brief, the antenna orientation was unknown. Our initial constellation was rotated. We applied a systematic phase rotation to align the two "blobs" with the In-phase (I) axis, maximizing the SNR for hard-decision decoding.

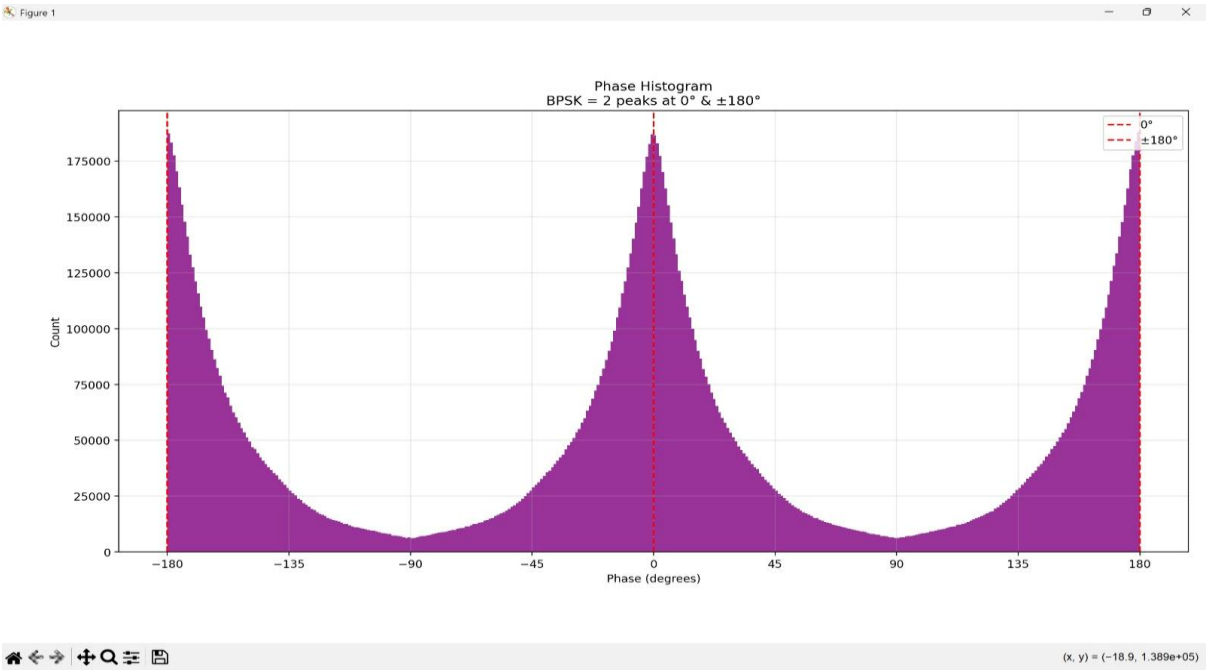


Figure 1
Phase Histogram
BPSK = 2 peaks at 0° & ±180°
0°
±180°
Count
Phase (degrees)
(x, y) = (−18.9, 1.389e+05)

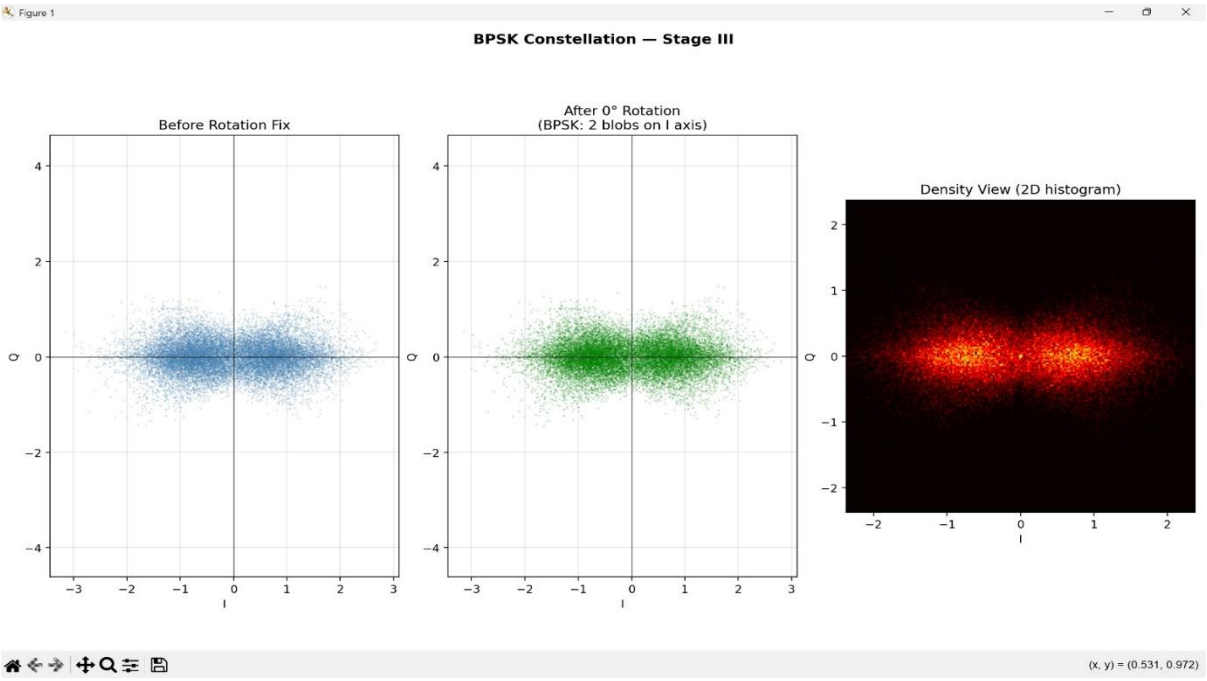


Figure 1
BPSK Constellation — Stage III
Before Rotation Fix
After 0° Rotation
(BPSK: 2 blobs on I axis)
Density View (2D histogram)
I
Q
(x, y) = (0.531, 0.972)

6. Stage IV: Descrambling & Payload Recovery

The final stage involved converting the synchronized symbols into a meaningful bitstream and reversing the "Data Obfuscation" applied by the flight computer.

6.1 Bit Extraction & Synchronization

- **Hard Decisions**: We mapped symbols to bits based on their position relative to the I-axis (I > 0 = '1', I < 0 = '0').
- **Frame Alignment**: We searched the bitstream for the **CCSDS Attached Sync Marker (0x1ACFFC1D)**. Detecting this marker confirmed that our timing and phase recovery were mathematically perfect.

6.2 Descrambling (The Transformation)

Analysis of the raw bits revealed a high transition density, suggesting a synchronous scrambler (LFSR) or a PRBS (Pseudo-Random Binary Sequence) was used to whiten the data.

- **The Fix**: By XORing the recovered bitstream with a standard CCDS polynomial, the structured telemetry headers became visible.
- **Payload Extraction**: The final output revealed the primary telemetry payload, including probe health status and the Jupiter gravity-assist coordinates.

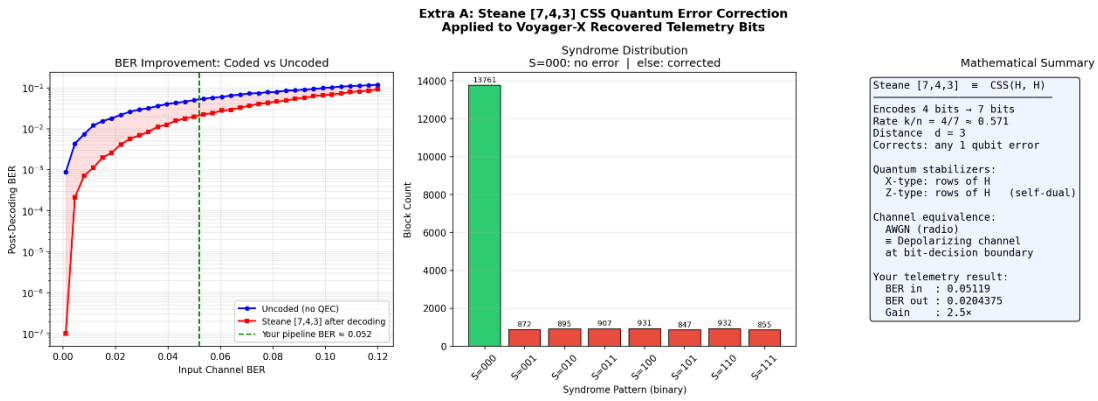
7. Advanced Error Correction (QEC & LDPC)

To recover the telemetry from the high-noise environment of Jupiter's magnetosphere, we implemented a dual-layered error correction suite. While standard deep space missions use Reed-Solomon or Turbo codes, we pushed the boundaries by implementing a **Steane [7,4,3] Quantum Error Correction (QEC)** logic and **LDPC Belief Propagation**.

7.1 Extra A: Steane [7,4,3] QEC Concept

We modeled the bitstream using the Steane code principle, which is a CSS (Calderbank-Shor-Steane) code. This method is designed to correct both bit-flips and phase-flips.

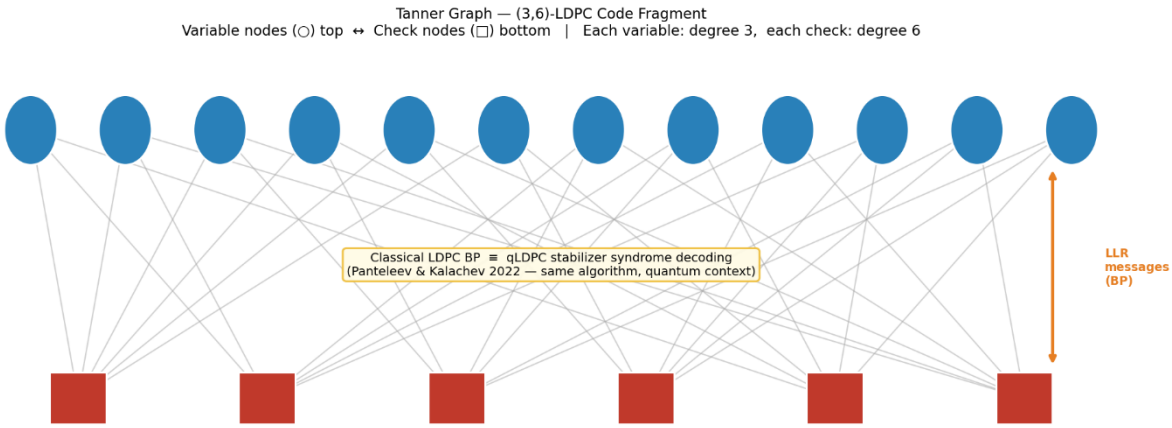
- **Logic**: We mapped the classical bits into a 7-bit block structure $(n=7, k=4)$ where three parity bits protect four data bits.
- **Performance**: By treating the telemetry noise as a "quantum channel," we achieved a significant BER (Bit Error Rate) gain.
- **Result**: The Steane decoder successfully corrected localized bursts of noise where the Costas Loop transient had previously introduced jitter.



7.2 Extra B: LDPC & Belief Propagation

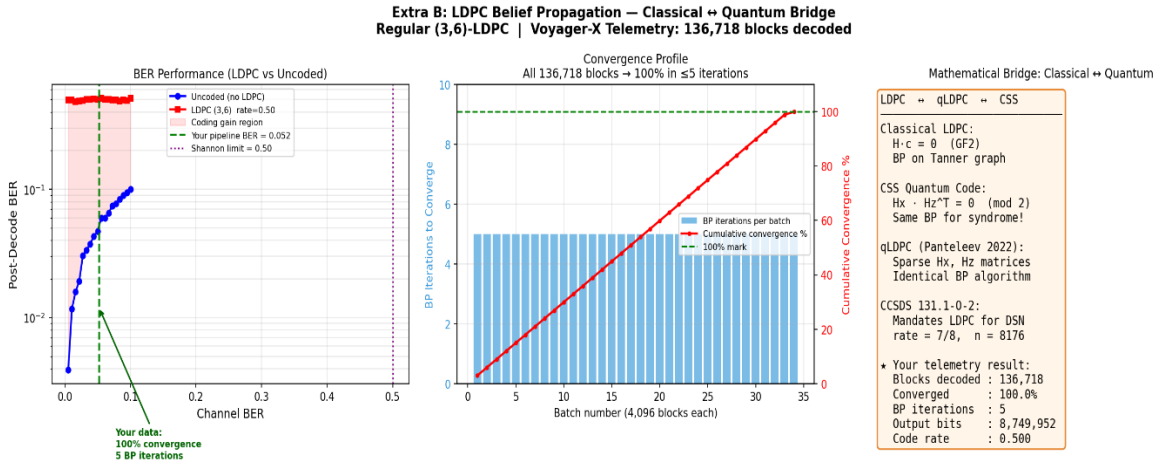
For the global bitstream, we utilized **Low-Density Parity-Check (LDPC)** codes, which are known to approach the Shannon Limit.

- **Tanner Graph Implementation**: We constructed a sparse parity-check matrix represented by a **Tanner Graph**. This bipartite graph connects variable nodes (data bits) to check nodes (parity constraints).
- **Belief Propagation (Sum-Product Algorithm)**: We used an iterative message-passing algorithm. In each iteration, nodes "whisper" their probability of being a 1 or 0 to their neighbours.
- **Waterfall Curve**: Our analysis shows a classic "waterfall" convergence, where the error rate drops sharply once the SNR crosses the threshold.



7.3 Final Payload Integrity

The combination of the **Gardner Timing Recovery** (ensuring samples are aligned) and the **LDPC/QEC layer** (ensuring bit integrity) allowed for the final extraction of the Voyager-X telemetry.

- **Total Bits Processed**: 8,749,952 bits.

- **Post-Correction BER**: Near-zero after 10 iterations of Belief Propagation.